Alberto Reyes Castro

Applied AI Engineer | Multimodal & On-Device ML | MSc AI Technology

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Profile

Physics graduate and MSc Artificial Intelligence Technology student at Northumbria University London, currently performing at Distinction level. Applied AI engineer focused on multimodal retrieval, efficient on-device inference, model evaluation and reliable AI systems. Builds systems end to end, from model selection and benchmarking through implementation, physical-device validation and human-review workflows. Seeking Summer 2027 machine-learning / Applied AI internships.

Experience

Applied AI Engineer - On-Device Multimodal ML | Jul 2026-Present
TeachScribe - Technical Collaboration, London / Hybrid | Full-time Jul-Oct; part-time since Oct

- Designed, built and evaluated a fully on-device multimodal photo-retrieval prototype for Android for TeachScribe, a UK Early Years platform in use in 50+ schools; local photos are ranked against written observations with a person confirming every match.
- Benchmarked five pretrained vision-language models on curated retrieval evaluations covering Hit@1/Hit@5, model size, CPU cost, mobile deployability and checkpoint licensing; selected TinyCLIP 39M/16 (17/20 Hit@1, 18/20 Hit@5 on a 20-query desktop set with 16 candidate images).
- Implemented INT8 ONNX Runtime inference with native Kotlin preprocessing, reusable sessions and a persistent SQLite embedding cache; dynamic INT8 quantisation reduced the text encoder from approximately 170 MiB to 43 MiB (74.8%).
- On a Huawei MediaPad M3 Lite 8 (Android 7), 108/108 cached searches over 44 photos completed at 441 ms median / 459 ms P95; lazy image sessions held idle memory at 76.3 MiB PSS versus 166.6 MiB with preloading. Source code and internal product materials remain private.

Python & Scientific Communication Intern | Mar 2025-May 2025
Instituto de Astrofisica de Canarias (IAC), Tenerife | 250h | 9.2/10

- Developed Python workflows for multilingual interview transcription/subtitling and semi-automated matching of 67 outreach articles against 700+ scientific publications using text preprocessing, regex and fuzzy matching with manual review for uncertain cases.
- Produced technical documentation and science communication in English and Spanish, including scientific interview material and published IAC outreach content.

Selected Projects

ModelHawk - ML Notebook Audit Prototype -- **GitHub**
Next.js/TypeScript tool that parses ML notebooks and applies deterministic and heuristic checks for leakage risk, metric choice, class imbalance, validation quality and reproducibility. Optional examiner notes use the Gemini API with a local Llama 3.2 fallback through Ollama; reports export to LaTeX/PDF.

Agent Control Center - Multi-Agent Orchestration -- **GitHub | Demo**
Public interactive simulation (six failure scenarios) of the control plane behind a private agent-orchestration harness, illustrating capability-aware routing, execution leases, bounded recovery, separate review, deterministic checks and a human merge gate. Authored demo data; no live telemetry or provider calls.

Credit Card Fraud Detection -- **GitHub**
MSc machine-learning project (85/100) on 284,807 transactions with leakage-controlled preprocessing, six classifiers, imbalance strategies, stratified cross-validation and validation-based thresholding. Tuned XGBoost: 0.8827 average precision, 91.2% precision and 84.7% recall.

Adaptive Whisper Transcription Pipeline -- **GitHub**
Reusable faster-whisper workflow with CPU/GPU-aware model selection, VAD, repetition safeguards, caching, quality checks and TXT/SRT export; evolved from scientific interview transcription work at the IAC.

AIoT Smart Home Prototype -- **GitHub**
ESP32-S3 / Edge Impulse / Wokwi system with profile-aware automation and deterministic safety fallbacks; simulated, rule-labelled held-out set: 85.9% accuracy, 0.99 AUC, 0.89 weighted F1.

IAC Publication-Press Note Matching -- **GitHub**
Semi-automated Python workflow comparing 67 outreach articles with 700+ scientific publications using preprocessing, regex and fuzzy matching with manual review.

Education

MSc Artificial Intelligence Technology with Advanced Practice | Jan 2026-Jan 2028
Northumbria University London, UK | Current

Currently performing at Distinction level, with first-semester marks including **85/100**, **75/100** and **70/100**. The 85/100 assessment was an end-to-end machine-learning project on credit-card fraud detection. Relevant study includes Programming for AI, Machine Learning on Cloud, Artificial Intelligence for IoT, and Information Governance & Cyber Security.

BSc Physics | Sep 2018-Jul 2025
Universidad de La Laguna, Tenerife, Spain | 7.13/10

Final Degree Project written and defended in English: structural mechanisms of ferroelectricity and ferroelasticity in beta'-RE2(MoO4)3 compounds, **9.4/10**. Academic Merit Convocation Award (2025).

Technical Skills

ML & evaluation: